

Practice Quiz: Cognition (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. The prefrontal cortex maintains working memory through: A) Permanent structural changes in neurons B) Sustained neural activity that persists after the stimulus is removed C) Storing information in neurotransmitter vesicles D) Rapid synapse formation and destruction
2. The dorsal attention network is: A) Automatic and stimulus-driven B) Goal-directed and voluntary — increasing precision on task-relevant information C) Only active during sleep D) Responsible for emotional processing
3. The P300 ERP component reflects: A) The onset of visual processing B) Context updating and belief revision following an unexpected stimulus C) Motor preparation D) Auditory processing exclusively
4. In the Active Inference account of schizophrenia, delusions arise from: A) Lack of intelligence B) Aberrantly high precision on prediction errors, making false beliefs feel certain C) Low brain activity overall D) Damage to the occipital lobe
5. Frontal theta oscillations (4-8 Hz) increase during: A) Relaxation B) Cognitive control, conflict detection, and belief updating C) Deep sleep only D) Repetitive motor tasks
6. The ventral attention network detects: A) Planned, anticipated stimuli B) Salient, unexpected stimuli that generate large prediction errors C) Only visual stimuli D) Internal thoughts
7. ADHD in the precision framework involves: A) Too much attention to everything B) Impaired precision modulation — difficulty boosting relevant and suppressing irrelevant signals C) Damage to the cerebellum D) Excessive serotonin

Part B: Short Answer

1. Explain how working memory in the PFC implements the approximate posterior $q(s)$ in Active Inference. What is being "held in mind," and why does it need to be updated?
2. A person with depression finds it hard to motivate themselves to do things they used to enjoy (anhedonia). Using the precision framework, explain which prediction errors are under-weighted and how this produces the symptom.
3. Compare top-down and bottom-up attention at the neural level. Give a real-life scenario where both operate simultaneously and explain how they cooperate.